

Image Processing

INT3404 20

Week 7: Feature extraction

Edge, Line, Texture

Lecturer: Nguyen Thi Ngoc Diep, Ph.D.

Email: ngocdiep@vnu.edu.vn

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Schedule

Tuần	Nội dung	Yêu cầu đối với sinh viên (ngoài việc đọc tài liệu tham khảo)
1	Giới thiệu môn học	Cài đặt môi trường: Python 3, OpenCV 3, Numpy, Jupyter Notebook
2	Ảnh số (Digital image) – Phép toán điểm (Point operations) Làm quen với OpenCV + Python Điều chỉnh độ tương phản (Contrast adjust)– Ghép ảnh (Combining images)	
3	Histogram - Histogram equalization	Làm bài tập 1: điều chỉnh gamma tìm contrast hợp lý
4	Phép lọc trong không gian điểm ảnh (linear processing filtering)	Thực hành ở nhà
5	Phép lọc trong không gian điểm ảnh (linear processing filtering) (cont.) Thực hành: Cách tìm filters	Thực hành ở nhà
6	Thực hành: Ứng dụng của histogram; Tìm ảnh mẫu (Template matching)	Bài tập mid-term
7	Trích rút đặc trưng của ảnh Cạnh (Edge) và đường (Line) và texture	Thực hành ở nhà
8	Các phép biến đổi hình thái (Morphological operations)	Làm bài tập 2: tìm barcode
9	Chuyển đổi không gian – Miền tần số – Phép lọc trên miền tần số Thông báo liên quan đồ án môn học	Đăng ký thực hiện đồ án môn học
10	Xử lý ảnh màu (Color digital image)	Làm bài tập 3: Chuyển đổi mô hình màu và thực hiện phân vùng
11	Các phép biến đổi hình học (Geometric transformations)	Thực hành ở nhà
12	Nhiều – Mô hình nhiễu – Khôi phục ảnh (Noise and restoration)	Thực hành ở nhà
13	Nén ảnh (Compression)	Thực hành ở nhà
14	Hướng dẫn thực hiện đồ án môn học	Trình bày đồ án môn học
15	Hướng dẫn thực hiện đồ án môn học Tổng kết cuối kỳ	Trình bày đồ án môn học

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Recall

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Week 7: Feature extraction

Edge, Line, Texture

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Edge detection

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What is an edge?

- An edge = a significant local change in the image intensity

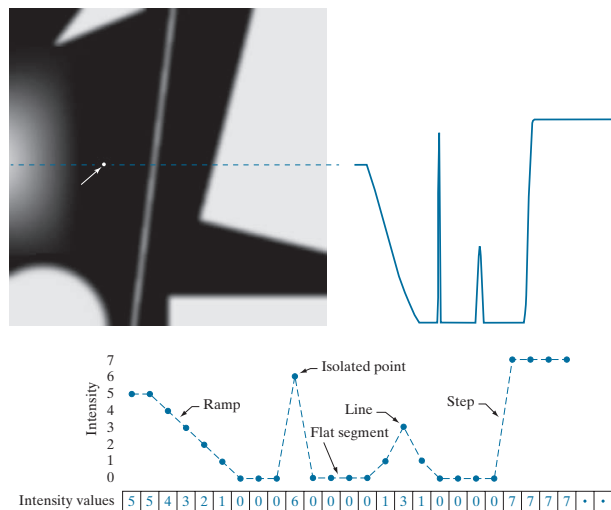
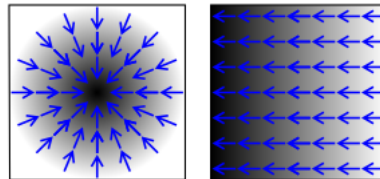


Image credit: Gonzalez et.al., Fig. 10.2

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Image gradient



Two types of gradients, with blue arrows to indicate the direction of the gradient. Dark areas indicate higher values

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Gradient

- Gradient of a function indicates **how strong the function increases**.

- For 1-dimension function: $f(x) = x^2$

$$\text{Grad}(x) = \frac{\partial f(x)}{\partial(x)} = 2x$$

- $\text{Grad}(2)=4$ indicates the the increasing direction of the function is to the right.
- $\text{Grad}(-1)=-2$ indicates the increasing direction of the function is to the left.

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First-order and second-order derivatives

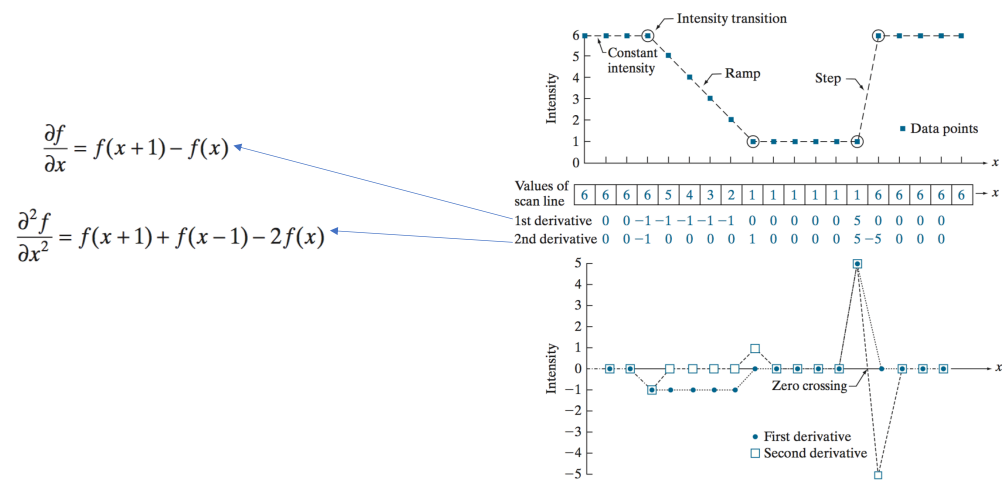


Image credit: Gonzalez, Fig. 3.44

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Edge detection using derivatives

(1) Detecting the **local maxima or minima** of the first derivative

(2) Detecting the **zero-crossings** of the second derivative

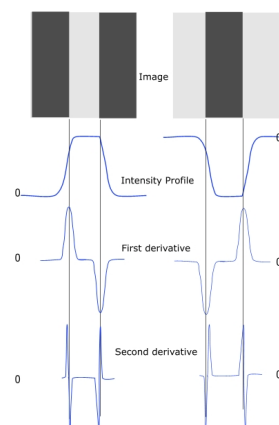


Image credit: MIPAV

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Gradient of 2D discrete function

- Gradient of a 2-dimension function is calculated as follows:

$$\text{Grad}(x, y) = \frac{\partial f(x, y)}{\partial x} \vec{i} + \frac{\partial f(x, y)}{\partial y} \vec{j}$$

- The gradient is approximated as follows (first-order derivative) :

$$\frac{\partial f(x, y)}{\partial x} = f(x+1, y) - f(x, y), \frac{\partial f(x, y)}{\partial y} = f(x, y+1) - f(x, y)$$

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Gradient of 2D image

$$\nabla f = \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

$$\frac{\partial f}{\partial y} = \begin{bmatrix} -1 \\ +1 \end{bmatrix} * \mathbf{A}$$

convolution

Approximated by finite differences



Avoid shifting of image

$$\begin{bmatrix} -1 \\ 0 \\ +1 \end{bmatrix}$$

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Gradient

- The magnitude of gradient indicates the strength of edges:

$$|Grad(x, y)| = \sqrt{\left(\frac{\partial f(x, y)}{\partial y}\right)^2 + \left(\frac{\partial f(x, y)}{\partial x}\right)^2}$$

- Gradient computation procedure:
 - Calculate column gradient
 - Calculate row gradient
 - Calculate final gradient by the above function

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Various kernels used to compute the gradient

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

A 3x3 region of an image

-1	0	0	-1
0	1	1	0

Roberts

-1	-1	-1	-1	0	1
0	0	0	-1	0	1
1	1	1	-1	0	1

Prewitt

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

Sobel

$$g_x = \frac{\partial f}{\partial x} = (z_9 - z_5)$$

$$g_y = \frac{\partial f}{\partial y} = (z_8 - z_6)$$

$$g_x = \frac{\partial f}{\partial x} = (z_7 + z_8 + z_9) - (z_1 + z_2 + z_3)$$

$$g_y = \frac{\partial f}{\partial y} = (z_3 + z_6 + z_9) - (z_1 + z_4 + z_7)$$

$$g_x = \frac{\partial f}{\partial x} = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$$

$$g_y = \frac{\partial f}{\partial y} = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$$

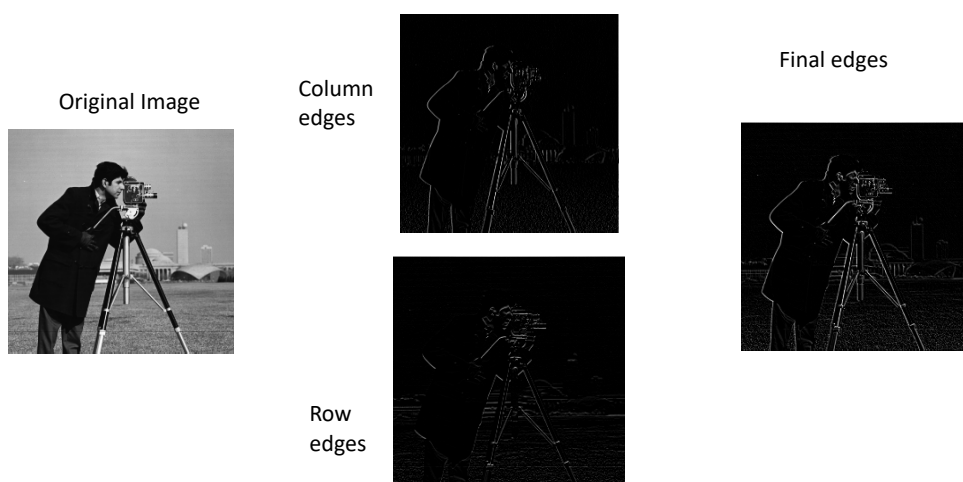
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Pixel Difference masks

Column mask	Row mask
$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

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Pixel difference example



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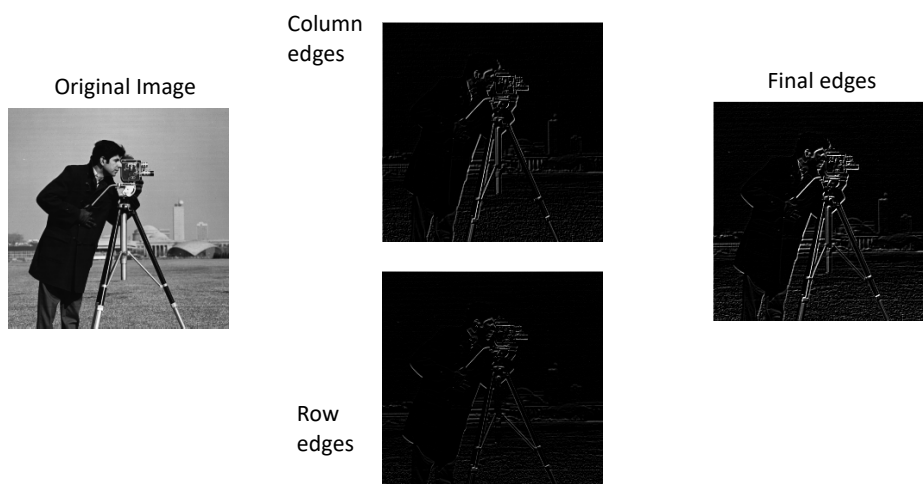
Robert mask

- Roberts masks calculate gradient from two diagonals

Column	Row
$\begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

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Robert mask example



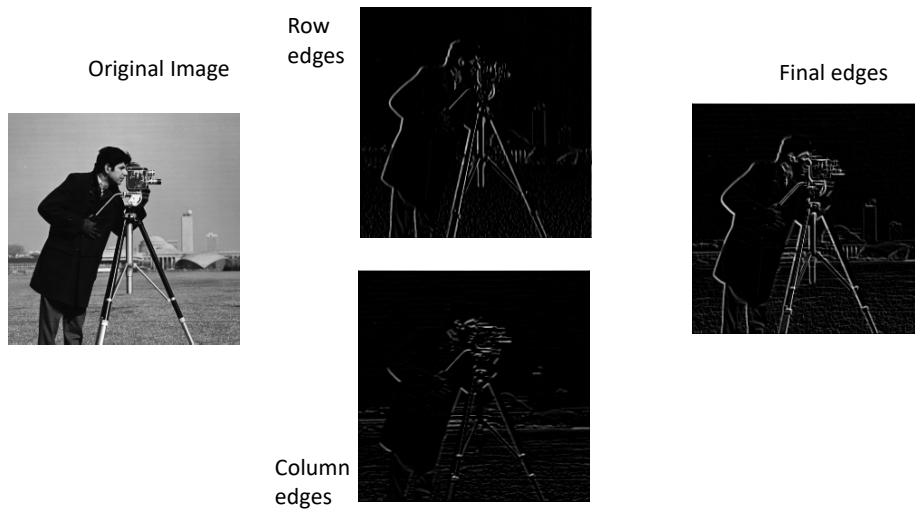
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Prewitt mask

$$\begin{array}{c} \text{Column} \\ \frac{1}{3} \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix} \end{array} \qquad \begin{array}{c} \text{Row} \\ \frac{1}{3} \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \end{array}$$

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Prewitt mask



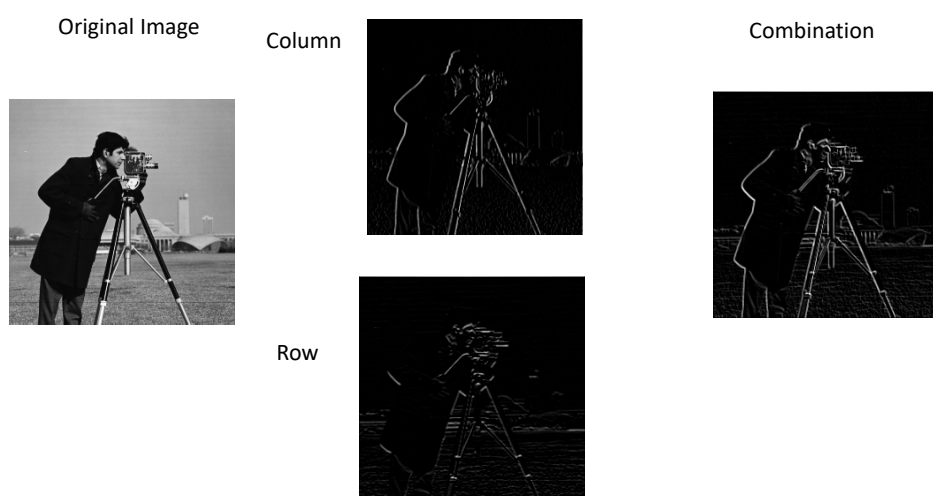
20

Sobel mask

$$\begin{array}{c} \text{Column} \\ \frac{1}{4} \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix} \end{array} \quad \begin{array}{c} \text{Row} \\ \frac{1}{4} \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \end{array}$$

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Sobel filter example



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Laplace gradient

- Laplace edge in a continuous domain

$$G(x, y) = -\left(\frac{\partial^2 f(x, y)}{\partial x^2} + \frac{\partial^2 f(x, y)}{\partial y^2}\right)$$

- In a discrete domain, Laplace edge is approximated by

$$\begin{aligned} G(x, y) &= [f(x, y) - f(x, y-1)] - [f(x, y+1) - f(x, y)] \\ &\quad + [f(x, y) - f(x+1, y)] - [f(x-1, y) - f(x, y)] \\ &= f(x, y) * H(x, y) \end{aligned}$$

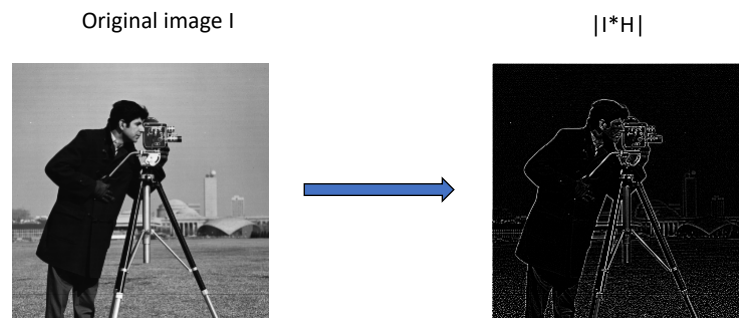
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Laplace mask

$$\begin{aligned} H &= \begin{bmatrix} 0 & 0 & 0 \\ -1 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & -1 & 0 \\ 0 & 2 & 0 \\ 0 & -1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix} \end{aligned}$$

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Laplace filter example



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Highboost filtering with Laplace

- Overall

$$I_{highboost} = c \cdot I_{original} + I_{highpass}$$

$$= \left(c \cdot \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + H \right) * I_{original}$$

- Using Laplace mask

$$I_{highboost} = \left(c \cdot \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix} \right) * I_{original} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & c+4 & -1 \\ 0 & -1 & 0 \end{bmatrix} * I_{original}$$

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Highboost filter example

Original Image



$c=0.5$



$c=1$



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Gradient comparison

Pixel difference



Robert



Prewitt



Sobel



Laplace



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More advanced edge detection

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LoG edge detection

1. Applying LoG to the image

$$\nabla^2 G(x, y) = \left(\frac{x^2 + y^2 - 2\sigma^2}{\sigma^4} \right) e^{-\frac{x^2 + y^2}{2\sigma^2}}$$

2. Detection of zero-crossings in the image
3. Threshold the zero-crossing to keep only the strong ones (large difference between the positive maximum and the negative minimum)

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LoG edge detection example



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Canny edge detection

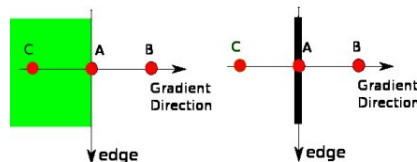
1. Smooth with 5x5 Gaussian kernel

2. Gradient with Sobel kernels

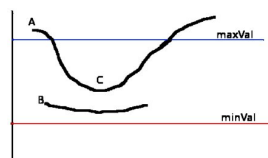
$$Edge_Gradient(G) = \sqrt{G_x^2 + G_y^2}$$

$$Angle(\theta) = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

3. Non-maximum suppression



4. Thresholding



<https://towardsdatascience.com/canny-edge-detection-step-by-step-in-python-computer-vision-b49c3a2d8123>

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Canny edge detection example

Original image



Canny



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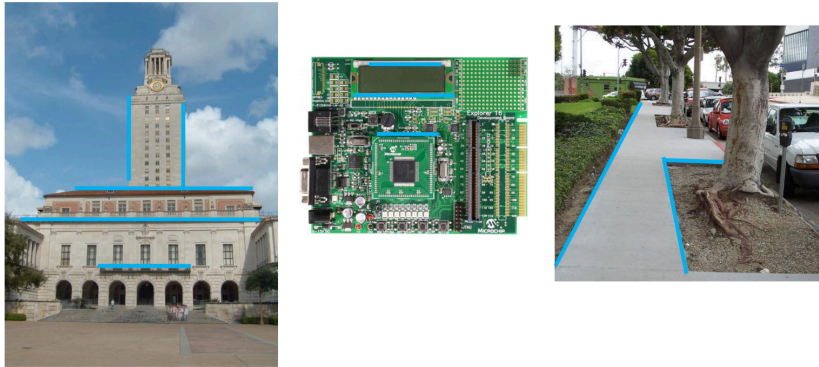
Line detection

Hough transform

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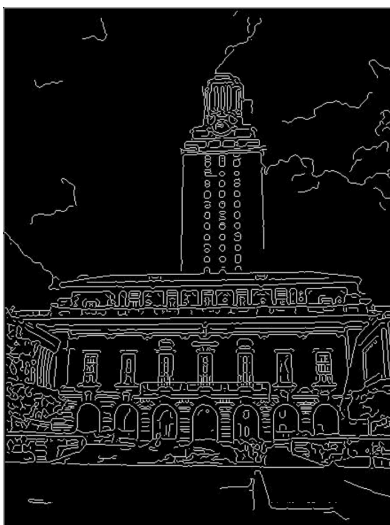
Example: Line fitting

- Many objects characterized by presence of straight lines



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Difficulty of line fitting

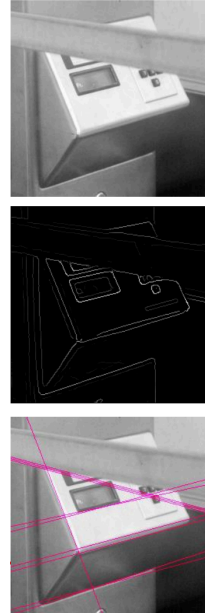


- **Extra** edge points (clutter), multiple models
 - Which points go with which line, if any?
- Only some parts of each line detected, and some parts are **missing**:
 - How to find a line that bridges missing evidence?
- **Noise** in measured edge points, orientations:
 - How to detect true underlying parameters

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Fitting lines with Hough transform

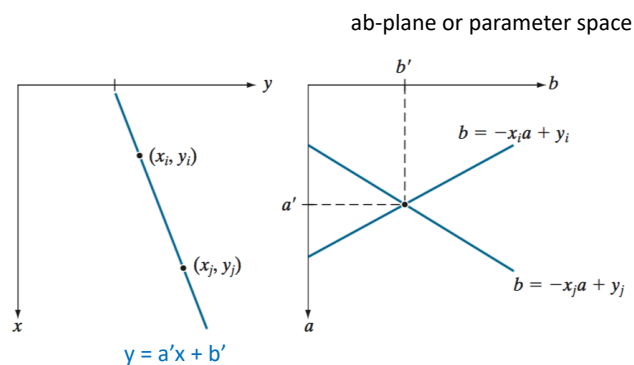
- Given points that belong to a line, what is the line?
- How many lines are there?
- Which points belong to which lines?
- Hough transform is a voting technique that can be used to answer all of these questions
- Main idea:
 1. Record vote for each possible line on which each edge point lies
 2. Look for lines that get many votes



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Line planes

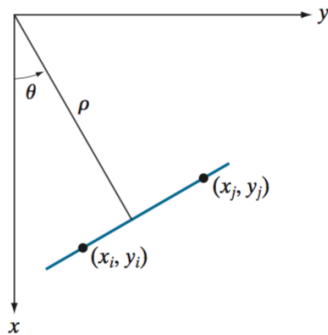
FIGURE 10.28
(a) xy -plane.
(b) Parameter space.



What if the line approaches the vertical or horizontal direction?
(i.e., infinity slope)

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Polar representation for lines



rho: perpendicular distance from line to origin

theta: angle the perpendicular makes with the x-axis

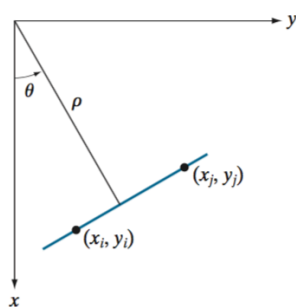
Normal presentation of a line:

$$x \cos \theta + y \sin \theta = \rho$$

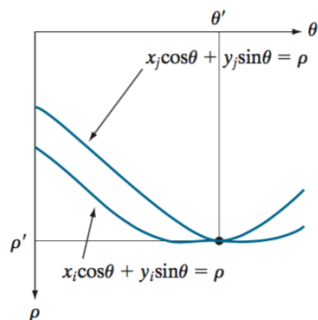
→ Point in image space is now sinusoid segment in Hough space

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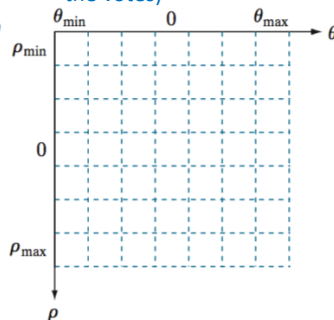
Finding lines in an image: Hough algorithm



1. Using the polar parameterization



2. Create a Hough Accumulator Array (keeps the votes)



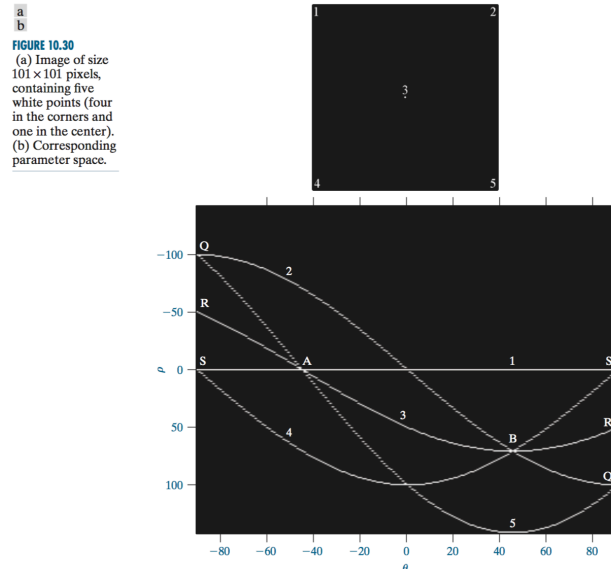
• Domain of the parametric space:

$$r \in [-\sqrt{M^2 + N^2}, \sqrt{M^2 + N^2}], \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

M and N image resolution

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Example of Hough transform



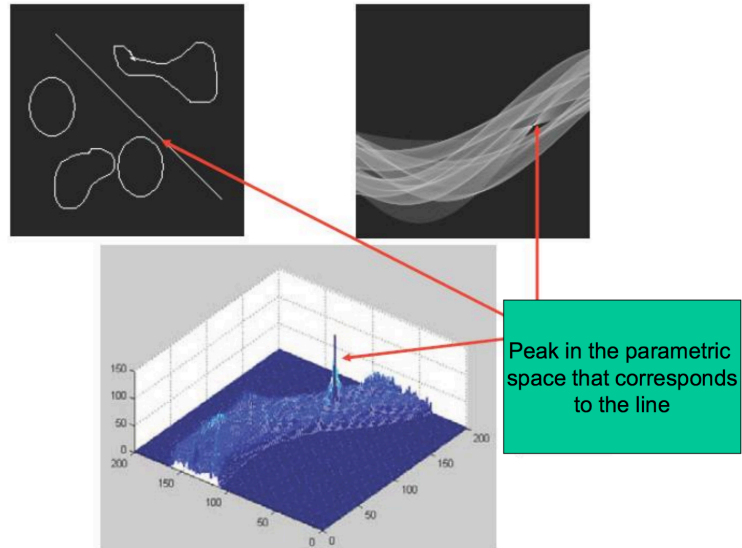
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Basic Hough transform algorithm

1. Initialize $H[d, \theta] = 0$
2. For each **edge** point in $E(x, y)$ in the image
 for $\theta = 0$ to 180 // some quantization; why not 2π ?
 $d = x \cos \theta + y \sin \theta$ // maybe negative
 $H[d, \theta] += 1$
3. Find the value(s) of (d, θ) where $H[d, \theta]$ is maximum
4. The detected line in the image is given
 by $d = x \cos \theta + y \sin \theta$

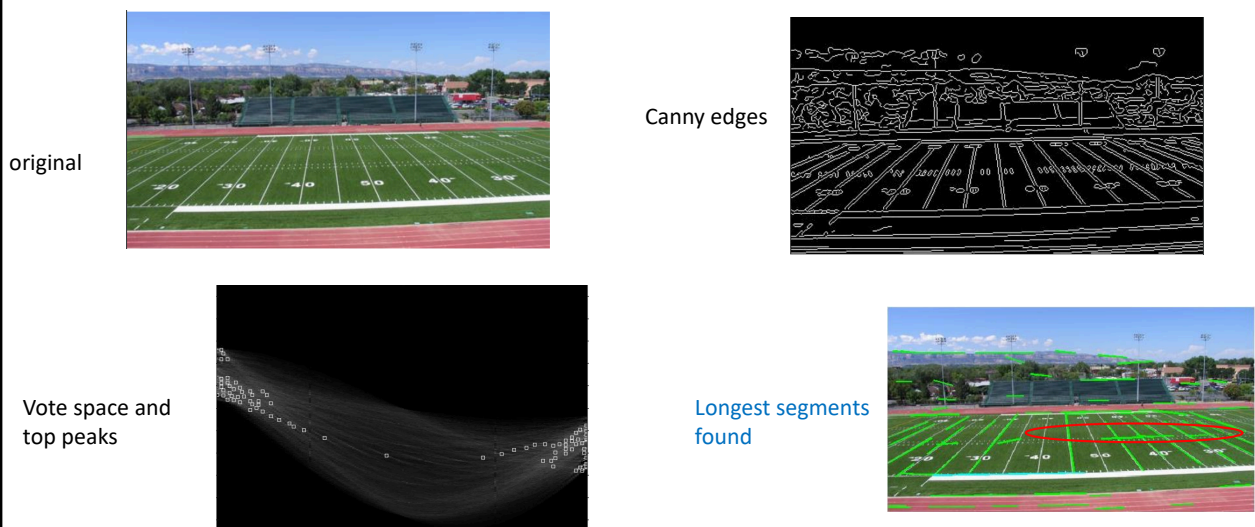
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Line detection example



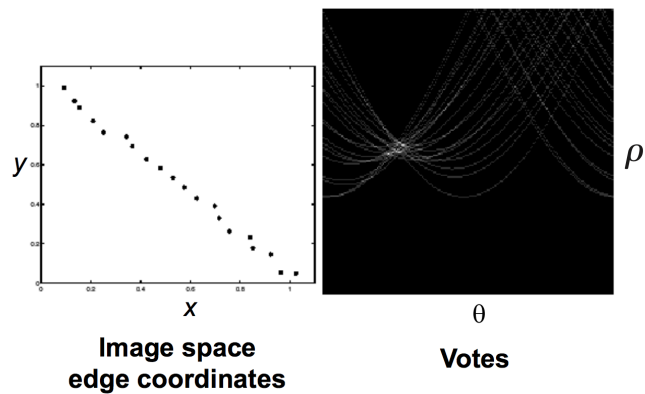
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Line detection example



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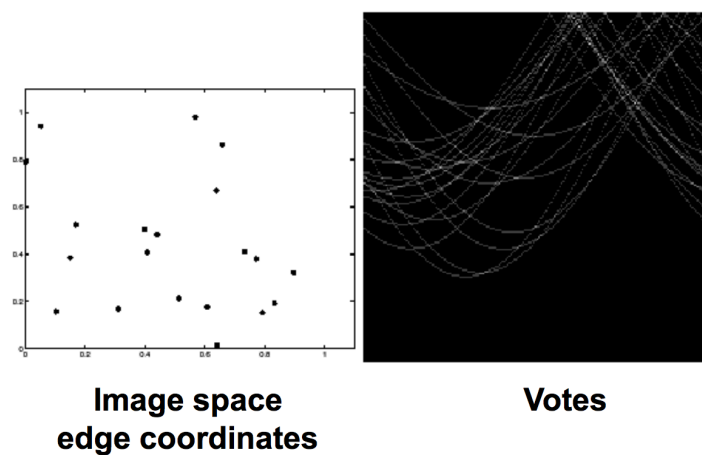
Impact of noise on Hough



What difficulty does this present for an implementation?

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Impact of noise on Hough



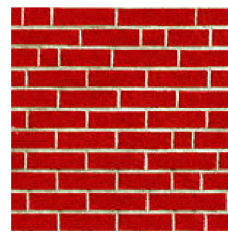
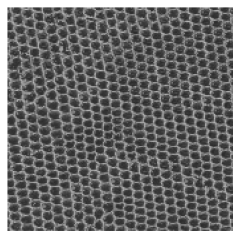
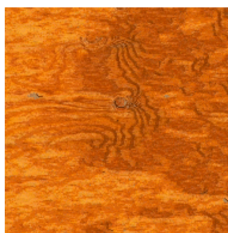
Everything appears to be “noise”, or random edge points, but we still see some peaks in the vote space

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Texture analysis

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What is texture?

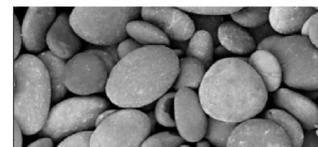
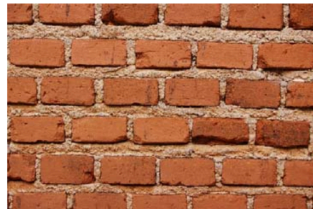


- An image obeying some statistical properties
- Similar structures repeated over and over again
- Often has some degree of randomness

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Aspects of texture

- Size/granularity (sand versus pebbles versus boulders)
- Directionality/Orientation
- Random or regular (stucco versus bricks)



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Statistical approach to texture

- Characterize texture using statistical measures computed from grayscale intensities (or colors) alone
- Less intuitive, but applicable to all images and computationally efficient
- Can be used for both classification of a given input texture and segmentation of an image into different regions

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Some (simple) statistical texture measures

- Edge density and direction
- Use an edge detector as the first step in texture analysis
- The number of edge pixels in a fixed-size region tells us how busy that region is
- The directions of the edges also help characterize the texture

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Two edge-based texture measures

1. edgeness per unit area

$$F_{\text{edgeness}} = |\{ p \mid \text{gradient_magnitude}(p) \geq \text{threshold} \}| / N$$

where N is the size of the unit area

2. edge magnitude and direction histograms

$$F_{\text{magdir}} = (H_{\text{magnitude}}, H_{\text{direction}})$$

where these are the normalized histograms of gradient magnitudes and gradient directions, respectively.

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Example

Original Image

Frei-Chen
Edge ImageThresholded
Edge ImageDifferent F_{edgeness} for different regions

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Local binary pattern measure

Multiresolution Gray Scale and Rotation Invariant Texture Classification with Local Binary Patterns

Timo Ojala, Matti Pietikäinen and Topi Mäenpää

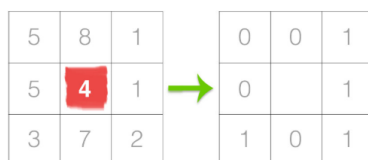


Figure 1: The first step in constructing a LBP is to take the 8 pixel neighborhood surrounding a center pixel and threshold it to construct a set of 8 binary digits.

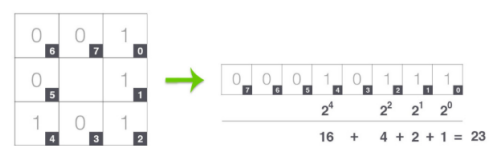


Figure 2: Taking the 8-bit binary neighborhood of the center pixel and converting it into a decimal representation. (Thanks to Bikramjot of [Hanzra Tech](#) for the inspiration on this visualization!)

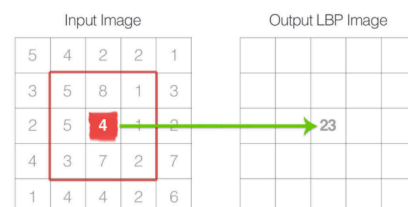
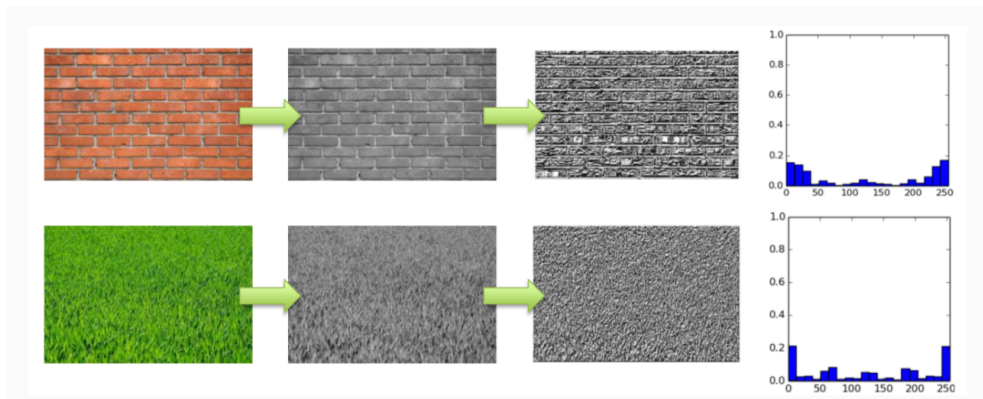


Figure 3: The calculated LBP value is then stored in an output array with the same width and height as the original image.

Image credit: <https://www.pyimagesearch.com/2015/12/07/local-binary-patterns-with-python-opencv/>

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LBP example



Color Image -> Grayscale Image -> LBP Mask -> Normalized LBP Histogram

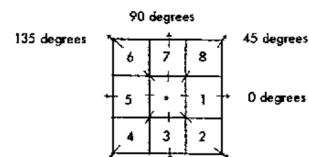
Image credit: <http://hanzratech.in/2015/05/30/local-binary-patterns.html>

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Gray Level Co-occurrence Matrix (GLCM) Textural Features for Image Classification

ROBERT M. HARALICK, K. SHANMUGAM, AND ITS'HAK DINSTEIN

- Distribution of co-occurring pixel values (grayscale values, or colors) at a given offset
 - A distance d , and an angle θ



Haralick features -> contrast, correlation texture features

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GLCM example

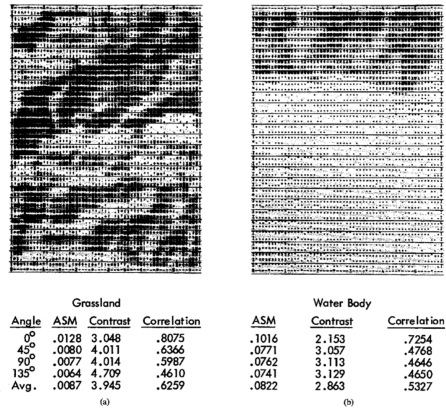


Fig. 4. Textural features for two different land-use category images.